Exercises: Convex Sets, Convex Functions, and Jensen's Inequality

Mathematical Foundations for Machine Learning

Convex Sets, Cones, and Convex Hulls

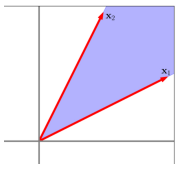
Exercise 1.1 [Easy]
Determine which of the following subsets of $\mathbb{R}^n$ are convex. Justify each answer.

(a) $S_1 = \{x \in \mathbb{R}^n \mid \|x\|_2 \leq 1\}$ (the closed unit ball).
(b) $S_2 = \{x \in \mathbb{R}^2 \mid x_1^2 + x_2^2 = 1\}$ (the unit circle).
(c) $S_3 = \{x \in \mathbb{R}^n \mid Ax \leq b\}$ for $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$ (a polyhedron).

Exercise 1.2 [Easy]
Let $K \subset \mathbb{R}^n$ be defined as $K = \{x \in \mathbb{R}^n \mid x_i \geq 0,\ i = 1, \ldots, n\}$ (the non-negative orthant).

(a) Show that K is a convex cone.
(b) Is K a proper cone? Justify your answer.

Pointed solid and closed
+ convex



sum inside
+scal inside

Exercise 1.3 [Medium]
Let $C_1, C_2 \subset \mathbb{R}^n$ be convex sets.

(a) Prove that $C_1 \cap C_2$ is convex.
(b) Give a counterexample showing that $C_1 \cup C_2$ need not be convex.
(c) The *Minkowski sum* is defined as $C_1 + C_2 = \{x + y \mid x \in C_1,\ y \in C_2\}$. Prove that $C_1 + C_2$ is convex.

Exercise 1.4 [Medium]
The convex hull of a set $S \subset \mathbb{R}^n$ is defined as

point in conv(S) = weighted combination of all points,
in boundary would be one single point with alfa = 1

$$\operatorname{conv}(S) = \Big\{ \sum_{i=1}^{k} \lambda_i x_i \;\Big|\; x_i \in S,\ \lambda_i \geq 0,\ \sum_{i=1}^{k} \lambda_i = 1,\ k \in \mathbb{N} \Big\}.$$

(a) Show that $\operatorname{conv}(S)$ is convex.
(b) Show that $\operatorname{conv}(S)$ is the smallest convex set containing S, i.e., if C is any convex set with $S \subseteq C$, then $\operatorname{conv}(S) \subseteq C$. z=(1k)w+kxk
(c) In the context of machine learning, suppose a neural network's weight space is $\mathbb{R}^d$ and we have a finite set of trained models $w_1, \ldots, w_k \in \mathbb{R}^d$. Describe geometrically what $\operatorname{conv}(\{w_1, \ldots, w_k\})$ represents. If the loss function $L(w)$ is convex, what can you say about the loss at any point in this convex hull?

$w = \sum_{i=1}^{k} \lambda_i w_i$

We take each weights as the average of all the models
that position conv({w1,…,wk}) -> if loss convex then bec.
of jensen the loss of the worst >= the hull's loss

Exercise 1.5 [Hard]
Let $f : \mathbb{R}^n \to \mathbb{R}$ be a convex function. Define the *sub-level set*

$$L_\alpha(f) = \{x \in \mathbb{R}^n \mid f(x) \leq \alpha\}.$$

(a) Prove that $L_\alpha(f)$ is a convex set for every $\alpha \in \mathbb{R}$.
(b) Give a counterexample showing that the converse is false: a function can have all sub-level sets convex yet fail to be convex.

-e^x is non convex (hessian negative)

(c) Let $f_1, \ldots, f_m : \mathbb{R}^n \to \mathbb{R}$ be convex functions and define $g(x) = \max_{i=1,\ldots,m} f_i(x)$. Prove that g is convex. Then show that the feasible set

$$\mathcal{F} = \{x \in \mathbb{R}^n \mid f_i(x) \leq 0,\ i = 1, \ldots, m\}$$

is convex.

(d) In the context of training a neural network with m constraints on fairness metrics, each of the form $f_i(w) \leq 0$ where f_i is convex, explain why the feasible region of model parameters is convex even though the intersection of constraints may be complex.

Convex and Concave Functions — Characterizations

Exercise 2.1 [Easy]
Classify each of the following functions as convex, strictly convex, strongly convex, concave, or none. Justify each answer using the second-order condition where applicable.

(a) $f(x) = 3x^2 + 2x + 1$ on $\mathbb{R}$.
(b) $g(x) = -\log(x)$ on $\mathbb{R}_{++}$.
(c) $h(x_1, x_2) = x_1 x_2$ on $\mathbb{R}^2$.

Exercise 2.2 [Easy]
Let $f : \mathbb{R}^n \to \mathbb{R}$ be convex and $g : \mathbb{R}^n \to \mathbb{R}$ be concave.

(a) Show that $h(x) = f(x) - g(x)$ is convex.
(b) Is $-f$ concave? Prove or disprove.
(c) Let $\alpha \geq 0$. Show that αf is convex.

Exercise 2.3 [Medium]
Let $f : \mathbb{R}^n \to \mathbb{R}$ be twice continuously differentiable.

(a) State the first-order and second-order sufficient conditions for f to be convex.
(b) Using the second-order condition, prove that $f(x) = \|Ax - b\|_2^2$ is convex for any $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$. Under what condition on A is f strongly convex? Intereseting eigenvalues proof
(c) In Ridge regression, the objective is

$$L(\theta) = \|X\theta - y\|_2^2 + \lambda \|\theta\|_2^2$$

Adding L2 regularization makes it strongly convex no matter X

for $\lambda > 0$. Show that L is 2λ-strongly convex regardless of X.

Exercise 2.4 [Medium]
Let $f : \mathbb{R}^n \to \mathbb{R}$ be convex and differentiable.

(a) Prove the first-order characterization: f is convex if and only if

$$f(y) \geq f(x) + \nabla f(x)^T (y - x), \quad \forall\, x, y \in \mathbb{R}^n.$$

(Hint: For the forward direction, use the definition of convexity with parameter $t \in (0, 1]$ and $z = x + t(y - x)$, then take $t \to 0^+$.)

(b) Use (a) to show that if f is convex and differentiable, then any critical point (where $\nabla f(x^*) = 0$) is a global minimizer.

Exercise 2.5 [Hard]
(a) Let $f : \mathbb{R}^n \to \mathbb{R}$ be m-strongly convex ($m > 0$). Prove that f has a unique global minimizer.

(Hint: Use the strong convexity inequality $f(y) \geq f(x) + \nabla f(x)^T (y - x) + \frac{m}{2}\|y - x\|^2$ to show existence of a lower bound, and then prove uniqueness by contradiction.)

(b) Let f be strictly convex on a convex set $C \subseteq \mathbb{R}^n$. Show that f has at most one minimizer over C. Give an example showing that a minimizer need not exist (i.e., strict convexity does not guarantee existence without further assumptions).

(c) Consider the negative log-likelihood in logistic regression:

$$L(w) = \sum_{i=1}^{N} \Big[\log\big(1 + \exp(w^T x_i)\big) - y_i\, w^T x_i \Big],$$

where $y_i \in \{0, 1\}$ and $x_i \in \mathbb{R}^d$. Show that L is convex. Is it strongly convex in general? What common regularization technique makes the objective strongly convex, and why?

Exercise 2.6 **[Hard]**
Let $f_1, \ldots, f_m : \mathbb{R}^n \to \mathbb{R}$ be convex functions, and let $A \in \mathbb{R}^{n\times p}$, $b \in \mathbb{R}^n$.

(a) Prove that $g(x) = \sum_{i=1}^{m} \alpha_i f_i(x)$ is convex for any $\alpha_i \geq 0$ (non-negative weighted sum preserves convexity).

(b) Prove that $h(y) = f_1(Ay + b)$ is convex in y (composition with affine mapping preserves convexity).

(c) Prove that $\phi(x) = \max_{i=1,\ldots,m} f_i(x)$ is convex (pointwise maximum preserves convexity).

(d) Give a counterexample showing that the composition $f_1 \circ f_2$ of two convex functions is not necessarily convex. What additional condition on f_1 makes it work?

Jensen's Inequality and Applications

Exercise 3.1 **[Easy]**
Jensen's inequality states: if f is convex and X is a random variable with $\mathbb{E}[|X|] < \infty$, then

$$f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)].$$

(a) Using $f(t) = t^2$, derive the inequality $\mathrm{Var}(X) \geq 0$ from Jensen's inequality.

(b) Using $f(t) = -\log(t)$ for $t > 0$, show that for positive values $a_1, \ldots, a_n$:

$$\frac{1}{n}\sum_{i=1}^{n} a_i \geq \left(\prod_{i=1}^{n} a_i\right)^{1/n}$$

(the AM-GM inequality).

Exercise 3.2 **[Medium]**
Let X be a random variable taking values in a convex set $C \subseteq \mathbb{R}^n$, and let $f : C \to \mathbb{R}$ be a **concave** function.

(a) State and prove Jensen's inequality for concave functions.

(b) Show that for any non-negative random variable X with finite mean:

$$\mathbb{E}[\sqrt{X}] \leq \sqrt{\mathbb{E}[X]}.$$

(c) In Bayesian machine learning, the *marginal log-likelihood* is $\log p(y) = \log \int p(y \mid w)p(w)\, dw$. Use Jensen's inequality applied to log (which is concave) to derive the *Evidence Lower Bound* (ELBO): for any distribution $q(w)$,

$$\log p(y) \geq \mathbb{E}_{q(w)}\left[\log \frac{p(y \mid w)p(w)}{q(w)}\right].$$

Exercise 3.3 [Hard]
(a) Let p and q be probability distributions over a finite set $\{1, \ldots, n\}$. The KL divergence is defined as

$$D_{\mathrm{KL}}(q\|p) = \sum_{i=1}^{n} q_i \log \frac{q_i}{p_i}.$$

Using Jensen's inequality applied to the convex function $f(t) = -\log(t)$, prove that $D_{\mathrm{KL}}(q\|p) \geq 0$ with equality if and only if $q = p$.

(b) Now consider the optimization problem

$$\min_q \; D_{\mathrm{KL}}(q\|p), \quad \text{s.t.} \quad q_i \geq 0, \; \sum_{i=1}^{n} q_i = 1.$$

Using part (a), what is the minimizer? Relate this to the entropy maximization problem

$$\max_q \; H(q) = -\sum_{i=1}^{n} q_i \log q_i, \quad \text{s.t.} \quad q_i \geq 0, \; \sum_{i=1}^{n} q_i = 1,$$

when p is the uniform distribution $p_i = 1/n$.

Epigraph, Subdifferential, Normal Cone, and Extreme Points

Exercise 4.1 [Medium]
(a) The *epigraph* of a function $f : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ is defined as

$$\operatorname{epi} f = \{(x, t) \in \mathbb{R}^{n+1} \mid f(x) \leq t\}.$$

Prove that f is convex if and only if $\operatorname{epi} f$ is a convex set.

(b) Consider the *extended-value* function $f : \mathbb{R} \to \mathbb{R} \cup \{+\infty\}$ defined by $f(x) = -\sqrt{x}$ for $x \geq 0$ and $f(x) = +\infty$ for $x < 0$. Sketch $\operatorname{epi} f$ and determine whether f is convex.

Exercise 4.2 [Medium]
(a) Compute the subdifferential $\partial f(x)$ of $f(x) = |x|$ at $x = 0$, $x = 2$, and $x = -3$.

(b) Let $f(x) = \max(0, 1 - x)$ (the hinge loss used in SVMs). Compute $\partial f(x)$ for all $x \in \mathbb{R}$.

(c) State and verify the optimality condition $0 \in \partial f(x^*)$ for the function $f(x) = |x-3| + 2|x|$. Find all minimizers.

Exercise 4.3 [Medium]
(a) Let $C = \{x \in \mathbb{R}^n \mid \|x\|_2 \leq r\}$ be the closed ball of radius r. Compute the normal cone $N_C(x)$ for:

- $x \in \operatorname{int}(C)$ (i.e., $\|x\|_2 < r$),
- $x \in \partial C$ (i.e., $\|x\|_2 = r$).

(b) A point x in a convex set C is an *extreme point* if x cannot be written as $x = \lambda y + (1-\lambda) z$ for $y, z \in C$ with $y \neq z$ and $\lambda \in (0, 1)$. Find all extreme points of:

- The simplex $\Delta = \{x \in \mathbb{R}^n \mid x_i \geq 0, \; \sum x_i = 1\}$.
- The ℓ_∞-ball $B_\infty = \{x \in \mathbb{R}^2 \mid \|x\|_\infty \leq 1\}$.

Mixed Challenging Exercises

Exercise 5.1 [Hard]
Consider the *log-sum-exp* function $f : \mathbb{R}^n \to \mathbb{R}$ defined as

$$f(x) = \log\left(\sum_{i=1}^{n} e^{x_i}\right).$$

(a) Prove that f is convex using Jensen's inequality.
(Hint: Interpret f as $\log \mathbb{E}[e^X]$ for an appropriate discrete random variable.)
(b) Show that $\max_i x_i \leq f(x) \leq \max_i x_i + \log n$. (This shows that log-sum-exp is a smooth approximation to the max function.)
(c) Compute the gradient $\nabla f(x)$ and the Hessian $\nabla^2 f(x)$. Show that $\nabla^2 f(x) \succeq 0$, giving an alternative proof of convexity.
(Hint: The Hessian can be written as $\mathrm{diag}(s) - ss^T$ where $s = \mathrm{softmax}(x)$. Use Cauchy-Schwarz to show $v^T(\mathrm{diag}(s) - ss^T)v \geq 0$.)

Exercise 5.2 [Hard]
Consider the following optimization problem from statistical learning. Let $p \in \mathbb{R}^n$ with $p_i > 0$ and $\sum p_i = 1$ be a fixed probability distribution. Solve:

$$\min_{x \in \mathbb{R}^n} \sum_{i=1}^{n} x_i \left(\log x_i - \log p_i\right), \quad \text{s.t.} \quad x_i \geq 0, \ \sum_{i=1}^{n} x_i = 1.$$

(a) Show that the objective function $\phi(x) = \sum_i x_i(\log x_i - \log p_i)$ is strictly convex on the simplex $\{x \mid x_i > 0, \sum x_i = 1\}$.
(b) Write the Lagrangian with multiplier μ for the equality constraint and multipliers λ_i for the inequality constraints $-x_i \leq 0$.
(c) Apply the KKT conditions to find the optimal solution x^*.
(d) Interpret the result: what does it say about when the KL divergence $D_{\mathrm{KL}}(x\|p)$ is minimized?

Exercise 5.3 [Hard]
Let $A \in \mathbb{R}^{n \times n}$ be symmetric positive semidefinite and $b \in \mathbb{R}^n$.
(a) Consider the set

$$\mathcal{E} = \{x \in \mathbb{R}^n \mid x^T A x + b^T x \leq 1\}.$$

Prove that $\mathcal{E}$ is a convex set.
(b) Let $f : \mathbb{R}^n \to \mathbb{R}$ be a strongly convex function with parameter $m > 0$. Prove that for any minimizer x^* and any $x \in \mathbb{R}^n$:

$$f(x) - f(x^*) \geq \frac{m}{2}\|x - x^*\|^2.$$

(c) A data scientist is training a model by minimizing $L(w) + \frac{\lambda}{2}\|w\|^2$ where L is convex but not strongly convex, and $\lambda > 0$. The training algorithm reaches a point $\hat{w}$ with $f(\hat{w}) - f(w^*) \leq \epsilon$. Using (b), derive an upper bound on $\|\hat{w} - w^*\|$ in terms of ϵ and λ.